

The Master Equation of SIGIL

A Conjectural Effective Action for a BlockDAG

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companion to *SIGIL — The Variational Chain*, whitepaper v1.1

Abstract

This note states the conjectural effective action that organizes the SIGIL protocol, defines its fields, maps each variation to a shipped or staged protocol component, and lists the falsifiable predictions the correspondence generates. **Its epistemic status is stated up front: this is a design principle and research programme, not accomplished mathematics.** The component Lagrangians are unspecified beyond their field content; no discretization of the BlockDAG into a lattice field theory has been constructed; each “field equation” below is a correspondence between a standard equation and a mechanism, adopted because it forces holistic design and yields testable predictions. One prediction generated under this discipline — the gossip delivery law — has completed the full arc from hypothesis to measured closed form. v0.2 incorporates corrections from external review: the PID stability discriminant, the gauge-healing language, the knot formulation, and the per-property statement of cryptographic composition.

1 The equation

SIGIL’s BlockDAG braid $\mathcal{G}$ — vertices are blocks, edges are causal links — is modeled as a Lorentzian 4-manifold. We conjecture its dynamics are organized by one variational principle:

$$\boxed{\delta \mathcal{S}_{\text{SIGIL}}[g, A, \varphi, J, K, \Sigma] = 0} \quad (1)$$

$$\mathcal{S}_{\text{SIGIL}} = \int_{\mathcal{G}} d^4\sigma \sqrt{|g|} \left[\frac{1}{2\kappa} (R - 2\Lambda_{\text{emission}}(\varphi)) + \mathcal{L}_{\text{settle}}(J, g) + \mathcal{L}_{\text{witness}}(A, g) + \mathcal{L}_{\text{topo}}(K) + \mathcal{L}_{\text{crypto}}(\Sigma, g) \right] \quad (2)$$

The claim this note is entitled to make: every SIGIL mechanism must identify which term it belongs to, and a proposed mechanism touching two or more terms is a design smell. The claim it is *not* yet entitled to make: that any mechanism has been derived from the action. The $\mathcal{L}$ ’s are names with declared field content, awaiting explicit form.

IN PLAIN WORDS — the one equation

Picture the whole chain as a stretchy fabric floating in time. Every block is a stitch; every fork is a wrinkle. The equation above is a single sentence claiming the fabric always settles into its least-wrinkled shape — money flowing, miners mining, watchdogs watching, even the cryptography, each one dial on the same machine. We have *not* proven the machine works this way. But designing *as if* it does keeps every piece pulling in the same direction — and it hands us predictions we can actually test.

2 The fields

Symbol	Type	Protocol meaning
$\mathcal{G}$	Lorentzian 4-manifold	The discrete BlockDAG braid, conjecturally modeled as a Lorentzian 4-manifold; the fabric of the chain.
$g_{\mu\nu}$	metric tensor	Conjectured to be induced by agent-reputation density; reputation as curvature. <i>Heuristic — functional form unfit.</i>
R	Ricci scalar	Consensus tension; forks raise it, convergence minimizes it.
$\Lambda_{\text{emission}}(\varphi)$	scalar field	The emission “cosmological constant,” PID-controlled toward target supply.
A_μ	gauge connection	Witness attestations; $F = dA + A \wedge A$ is the witness-disagreement field.
J^μ	conserved current	Settlement flow; J^0 value density, J^i settlement current.
K	link (closed braid)	The link obtained by closing the DAG’s braid over a bounded window; consensus-equivalent histories share its invariants.
$\Sigma = (\Sigma_{\text{iso}}, \Sigma_{\text{lat}}, \Sigma_{\text{hash}})$	security connection	Three disjoint post-quantum hardness assumptions, one per layer they guard.
$\kappa = 8\pi G_{\text{agent}}$	coupling	The gravitational constant of the agent economy; empirical.

3 The correspondences

Each variation names one protocol component. “Names,” not “yields”: see the abstract.

IN PLAIN WORDS — “correspondence, not derivation”

We are saying the protocol *rhymes* with famous physics — not that we’ve done the mathematics forcing it to. Think of a city map drawn in pencil: genuinely useful for navigating, honest about being redrawable. Each pencil line gets inked only when a measurement makes it earn the ink — one already has (Prediction 7, the delivery law).

3.1 $\delta\mathcal{S}_{\text{SIGIL}}/\delta J^\mu = 0$ — conservation (the ledger)

$$\nabla_\mu J^\mu = 0 \iff \partial_t \rho_{\text{value}} + \nabla \cdot \mathbf{J}_{\text{settle}} = S_{\text{emission}}(\varphi) \quad (3)$$

Value conservation with the emission source term. Implemented as the `commit_state_transition` chokepoint — the only code path permitted to mutate chain state — with per-delta conservation checks and the 21M cap as the integral form.

3.2 $\delta\mathcal{S}_{\text{SIGIL}}/\delta\varphi = 0$ — Klein–Gordon emission (monetary policy)

$$\ddot{\varphi} + 3H\dot{\varphi} + V'(\varphi) = 0, \quad V(\varphi) = K_p e^2 + K_i e \int e dt + K_d (\dot{e})^2 \quad (4)$$

The PID emission controller cast as a scalar field in a potential; Hubble friction is network expansion. The displayed V is *schematic*: e , $\int e dt$, and $\dot{e}$ are controller states, not functions of φ alone, so V is not an ordinary local potential — a faithful treatment introduces auxiliary fields (or defines $e(\varphi)$ explicitly). The mapping is exact only for quadratic potentials — anti-windup, derivative filtering, and gain scheduling break the analogy and are deferred to higher-order correction terms. **Corrected stability condition (v0.2):** from the linearized characteristic equation $K_d s^2 + K_p s + K_i = 0$, the non-oscillatory (overdamped) condition is

$$K_p^2 \geq 4 K_d K_i \quad (5)$$

v0.1 stated $K_d^2 > 4 K_p K_i$, interchanging the damping and inertia roles; a fresh derivation against the *implemented* controller (which is not exactly this second-order plant) is part of the prediction’s test protocol.

3.3 $\delta\mathcal{S}_{\text{SIGIL}}/\delta K = 0$ — topological invariance (fork detection)

$$\left. \frac{d}{dt} V_K(q) \right|_{\text{consensus}} = 0 \quad (6)$$

where K is the link obtained by closing the braid over a bounded window (v0.1’s $K \in \pi_1(\mathcal{G})$ was a category error — a knot is not an element of the fundamental group). Valid consensus moves are

Reidemeister moves; invariants are conserved; a fork *may* change the bounded-window invariant — when it does, divergence is proven, and equality of invariants never proves equality of histories. Two constraints bind the programme. *Computational*: the Jones polynomial is #P-hard in general, so the per-block commitment uses cheap invariants — linking number ($O(\text{crossings})$) or the Alexander polynomial (polynomial-time via the Burau determinant) — over a bounded window, with Jones/Khovanov machinery reserved for rare deep-fork adjudication. *Logical*: cheap invariants fingerprint, they do not classify. A differing invariant proves divergence; identical invariants do not prove identical histories. The topology commitment is a one-way tripwire that complements, never replaces, the state-root comparison.

3.4 $\delta\mathcal{S}_{\text{SIGIL}}/\delta A^\mu = 0$ — Yang–Mills witnesses (divergence detection)

$$\nabla_\nu F^{\mu\nu} = J_{\text{witness}}^\mu, \quad F = dA + A \wedge A \quad (7)$$

Diverging state-root tracks make the curvature F locally non-zero; detection is the measurement of $|F|$ above threshold. **Corrected language (v0.2)**: healing is *re-synchronization dynamics* that drives F back toward zero. It is not a gauge transformation — curvature is gauge-covariant, so a gauge transformation relabels the connection without repairing the underlying disagreement. In code: four state roots per header, independent recomputation at every node, self-halt on mismatch.

3.5 $\delta\mathcal{S}_{\text{SIGIL}}/\delta\Sigma = 0$ — crypto-agility (the security floor)

Three disjoint post-quantum hardness families, each guarding the layer where its strengths bind: Σ_{iso} (isogeny — SQIsign L5; identity and provenance, sign once, 292 B), Σ_{lat} (lattice — ML-DSA/ML-KEM; per-transaction and transport, fast batched verification), Σ_{hash} (FRI-STARK and the Ajtai/SIS fold; state and tip-proof, succinct whole-chain verification, no trusted setup). The correspondence motivates the dispatch rule: retire a leg the instant its hardness margin drops; the others carry what they guard. **Composition, stated carefully (v0.2)**: because the legs protect *different* properties, the security of any single property is the floor of the primitives guarding *that* property; an adversary targeting one property attacks one leg. The break-all-three bar applies only to forging a block’s complete commitment stack as a coherent whole, and a formal joint security model is open work.

3.6 The constraint surface — the wall

SIGIL imposes the engineering feasibility condition $\Gamma_{\text{verify}} \geq \Gamma_{\text{state}}$; we call the boundary “the wall.” It is an operational constraint, not a proven condition for the existence of an action extremum. Measured 2026-05-30 at $\sim 1840:1$ against verification, resolved not by faster per-signature verification but by changing Σ_{hash} : the fold collapsed whole-chain verification to a constant-size check (2,568 B; 342 ms full verification at the 100k-block benchmark; 3 μs live incremental tip check). The wall was a cryptographic choice, not a hardware limit — the one structural claim of this note that events have since borne out.

IN PLAIN WORDS — the wall

We assumed verification speed was a hardware problem — buy faster machines, verify faster. It turned out to be a *choice-of-mathematics* problem: switch the proof system, and checking the whole chain becomes one constant-size receipt instead of a million re-computations. The wall didn’t get climbed. It got deleted — exactly where the security term of the action said the pressure was.

4 Predictions

The correspondence earns its keep only by predicting. Status at v0.2:

Prediction	Status	Test
1. Fork survival $\propto \exp(-\Delta R \tau_{\text{block}})$	<i>open</i>	fork-survival statistics vs. producer reputation
2. PID non-oscillatory iff $K_p^2 \geq 4K_dK_i$ (corrected)	<i>open</i>	derive against implemented controller; tune to critical value
3. Bounded-window topology commitment detects order-tampering forks	partially met	braid gates: 16/16 permutations agree, 5/5 tamper rejected; header commitment staged
4. The verification wall scales with Σ , not hardware	confirmed in kind	the fold moved the wall; SIMD did not
5. Bulk/boundary duality (settlement throughput $\sim$ boundary entropy production)	<i>open</i>	compare vs. reputation-graph entropy
6. Validator-departure entropy law (Hawking analogue)	<i>open</i>	tip-proof rate vs. validator-set entropy
7. Gossip delivery follows $D(p, r) = 1 - p^r$	confirmed	22 seeded simulations, RMS 0.4 pp; shipped as the $r=3$ provisioning default

Prediction 7 is the template: hypothesis $\rightarrow$ seeded simulation campaign $\rightarrow$ closed form $\rightarrow$ residual $\rightarrow$ operator-facing rule $\rightarrow$ published paper. Predictions 1–6 are held to that arc.

5 What’s still pretend

In one place, so no reader has to hunt: (i) the action is conjectural — Lagrangians unspecified, no lattice discretization, correspondences not derivations; (ii) the reputation metric is a heuristic awaiting a fit to measured consensus drift; (iii) the tractable topology programme is a fingerprint, not a classifier, and the header commitment is staged; (iv) AdS/CFT on a discrete graph is analogy pending rigorous discretization; (v) the PID/Klein–Gordon mapping holds only for quadratic potentials; (vi) per-property crypto floors are design intent — the joint composition bound and the fold’s concrete Ajtai/SIS soundness reduction are open; (vii) the 1840:1 saddle measurement was taken on one 48-core host — the saddle condition is hardware-agnostic, its scale is not.

6 Method

This note was produced under the project’s measure-first discipline: the fields were chosen to match modules that already exist in the workspace (**flux-science**: Einstein–Hilbert, holographic, inflation, and quantum-gravity components; the **flux-topology** invariants; the fold), each correspondence was cross-checked against session measurements (settlement rates, state/verify throughput, the divergence event record), and every gap found in review was folded back into the text rather than argued away. The v0.2 corrections — discriminant, gauge language, knot formulation, composition statement — came from exactly that loop.

References

- *SIGIL — The Variational Chain*, whitepaper v1.1 (2026-07-18) — https://quillon.xyz/downloads/SIGIL_WHITEPAPER_v1.pdf
- *A Closed-Form Delivery Law for Gossip Redundancy in the sigil-top Light Node* (2026-06-17) — <https://quillon.xyz/downloads/sigil-top-delivery-law.pdf>
- *SIGIL $\times$ QTFT* (2026-05-30) — `SIGIL_QTFT_TOPOLOGY_v0.md`
- The SQInstructor: a guide to SQIsign and the Deuring Correspondence — arXiv:2603.09899
- High-Throughput GPU Implementation of Dilithium — arXiv:2211.12265
- A Comparative Analysis of zk-SNARKs and zk-STARKs — arXiv:2512.10020

v0.2 — 2026-07-18. The chain is modeled as a 4-manifold with curvature. The equation is conjecture; its discipline, and one of its predictions, are not.